

Problem Set 1

Applied Stats/Quant Methods 1

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Question 1: Education

Question 1.1: Confidence Interval

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
2       80, 97, 95, 111, 114, 89, 95, 126, 98)
3 # sample mean
4 mean_y <- sum(y)/length(y)
5 mean_y
6
7 # sample median
8 sorted_y <- sort(y, decreasing = FALSE)
9 sorted_y
10 median_y <- y[(length(y)%/%2)+1]
11 median_y
12
13 # sample variance
14 var_y <- (sum((y - mean_y)^2)/(length(y)-1))
15 var_y
16
17 # sample standard deviation
18 sd_y <- sqrt(var_y)
19 sd_y
20
21 # sample standard error
22 se_y <- sd_y/sqrt(length(y))
23 se_y
24
25 # assuming normal distribution of IQs, 90% CI:
26 # z-score for 90% CI normal distribution = 1.645
27
28 ci_90_lower <- mean_y - (1.645 * se_y)
```

```

29 ci_90_upper <- mean_y + (1.645 * se_y)
30
31 ci_90_lower
32 mean_y
33 ci_90_upper

```

This process results in a 90 percent confidence interval of [94.13, 102.75]

Question 1.2: Hypothesis Testing

Before beginning, it is important to note our assumptions, namely

1. That the data from the teacher's school is randomly sampled and therefore a good representation of the school population
2. That the nation-wide student IQs are normally distributed about the mean, 100.

Now, for the actual process:

1. First we establish the Null and Alternative Hypothesis:

```

1 # H0: country mean (mean_country = 100) >= sample mean (mean_y)
2 # HA: country mean (mean_country = 100) < sample mean (mean_y)

```

2. Next, we derive the test statistic:

```

1 test_statistic <- (mean_y - 100) / se_y

```

Which yields a value of -0.596.

3. As the sample size (n) is less than 30, we will reference the t-distribution and use pt() to find the p-value for this test statistic:

```

1 pt(abs(test_statistic), df = 24, lower.tail = TRUE)

```

Yielding p-value: 0.72

4. Finally, interpreting this result by comparing it to our alpha level ($\alpha = 0.05$):
Because 0.72 is greater than 0.05, we fail to reject the null hypothesis; we cannot say there is statistically significant evidence that this teacher's school's average IQ is above the national average school IQ.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

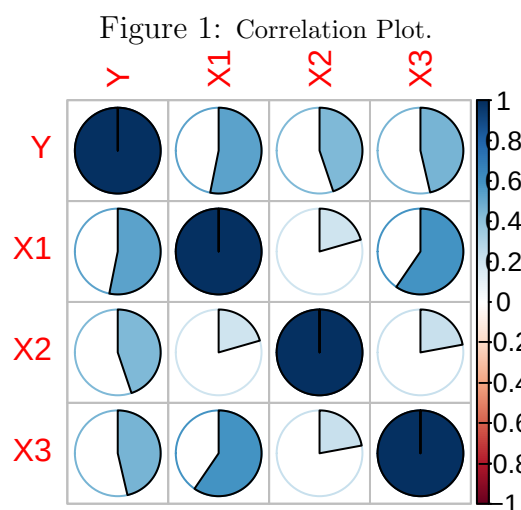
Question 2.1: Correlations

Using base R we can calculate correlations among the variables:

```
1 cor_expenditure <- cor(expenditure[, c("Y", "X1", "X2", "X3")])
```

This yields a matrix of correlations among the variables, which we can plot using package "corrplot":

```
1 corrplot(cor_expenditure, method = "pie")
```



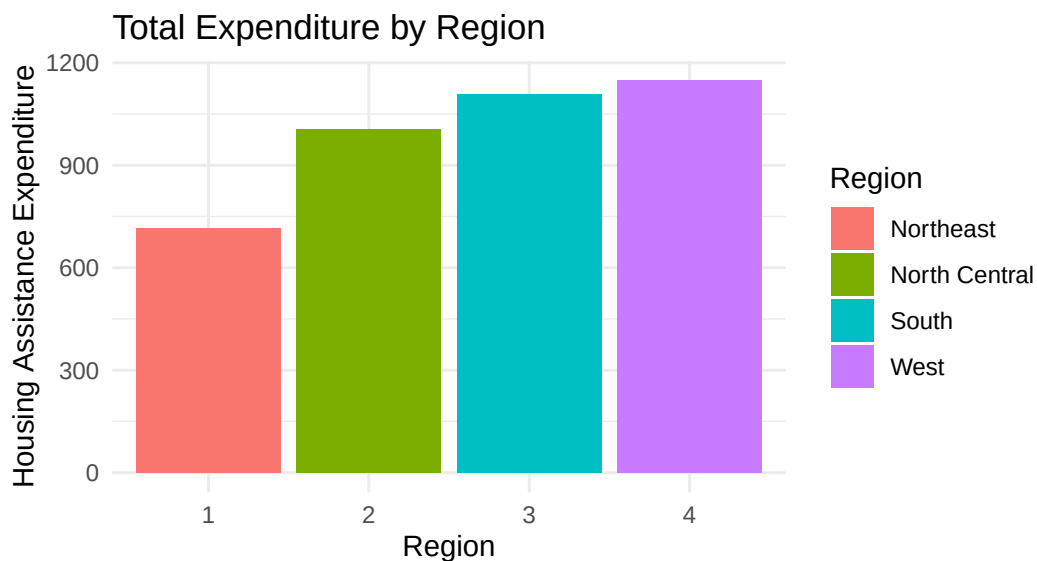
I chose method "pie" as this is most intuitive for me to visualize the different strengths of correlations between each variable pair. As we can see, the strongest correlating input variable for output variable Y (Housing Expenditure) is X1 (Per Capita Personal Income), whereas the weakest correlating variable is X2 (Number of residents per 100,000 that are "financially insecure" in state). We can also see the strength of the relationships between each of the input variables, strongest being between X1 and X3.

Question 2.2: Y vs. Region

Experimenting with the aggregate command in R, I created a plot of the total regional expenditures:

```
1 ?aggregate
2
3 regional_expenditure <- aggregate(Y ~ Region,
4                                   data = expenditure,
5                                   sum, na.rm = TRUE)
6 regional_expenditure
7
8 ggplot(regional_expenditure, aes(x = factor(Region),
9                                   y = Y,
10                                  fill = factor(Region)))+
11   geom_col() +
12   labs(x = "Region",
13         y = "Housing Assistance Expenditure",
14         title = "Total Expenditure by Region",
15         fill = "Region",
16   ) +
17   scale_fill_discrete(name = "Region",
18                       labels = c("Northeast",
19                                  "North Central",
20                                  "South",
21                                  "West"))
```

Figure 2: Total Regional Expenditures Plot.



This plot is fine for viewing the data, but it doesn't help us compare the regions— so we take the averages and compare those:

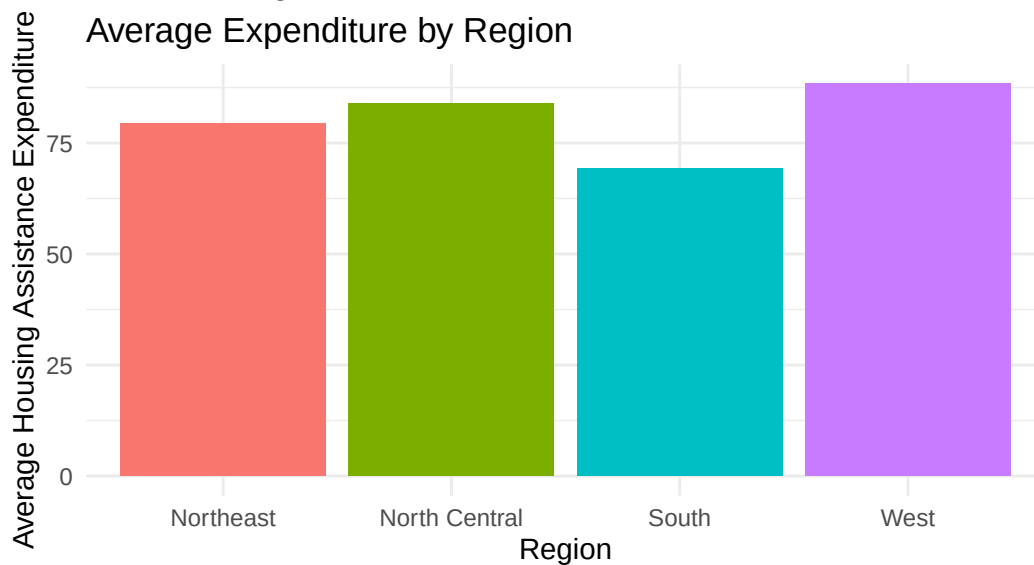
```
1 regional_averages <- aggregate(Y ~ Region,
```

```

2 data = expenditure ,
3 mean, na.rm = TRUE)

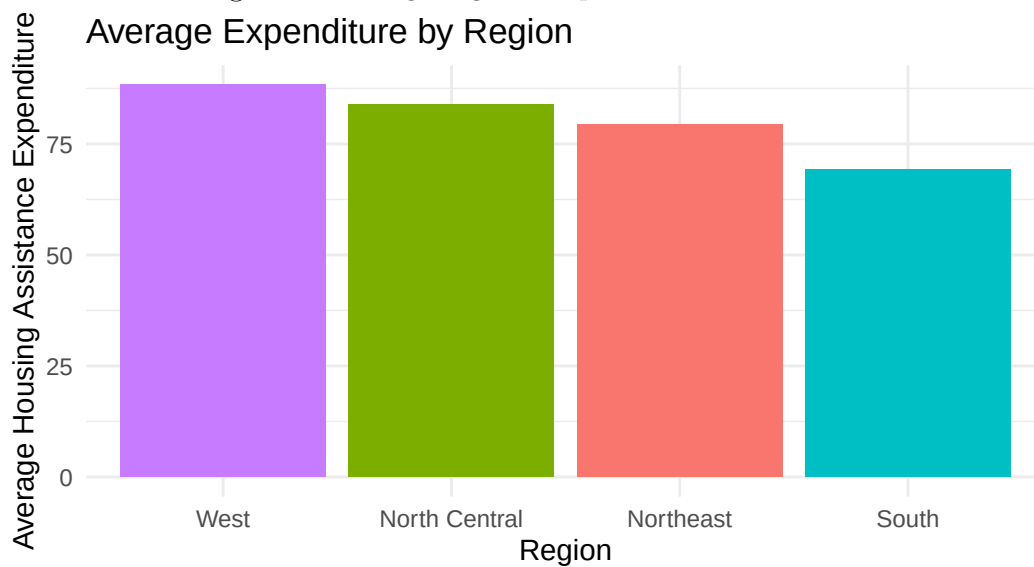
```

Figure 3: Average Regional Expenditures Plot.



And finally, it's helpful to view the graph in descending order:

Figure 4: Average Regional Expenditures Plot-Sorted.

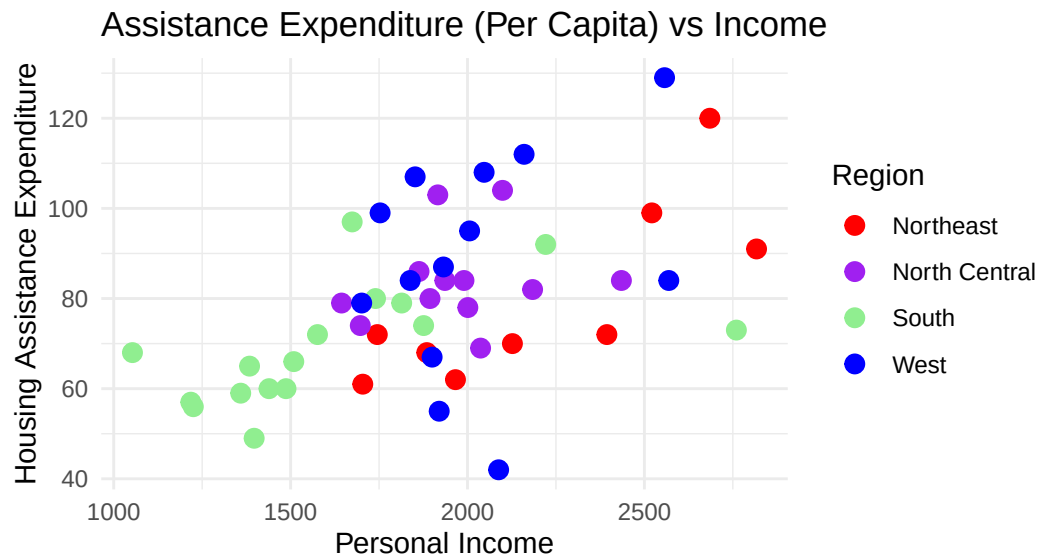


Question 2.3: Y vs. X1

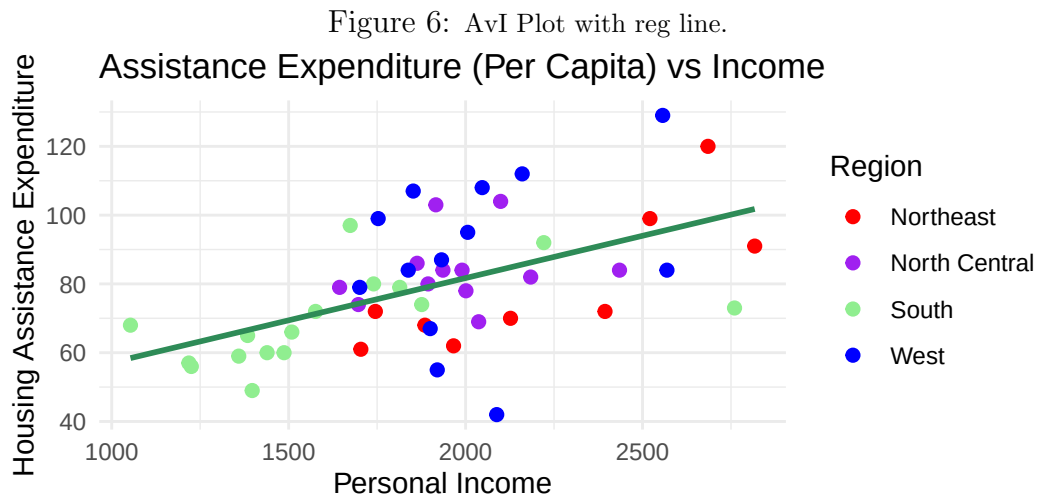
First, I took a general look at my data with a ggplot scatterplot:

```
1 ggplot(expenditure, aes(x = X1,  
2                        y = Y,  
3                        color = factor(Region))) +  
4   geom_point(size = 3) +  
5   scale_color_manual(values = c("red", "purple", "lightgreen", "blue"),  
6                        labels = c("1" = "Northeast",  
7                                   "2" = "North Central",  
8                                   "3" = "South",  
9                                   "4" = "West")) +  
10  labs(title = "Assistance Expenditure (Per Capita) vs Income",  
11        x = "Personal Income",  
12        y = "Housing Assistance Expenditure",  
13        color = "Region")
```

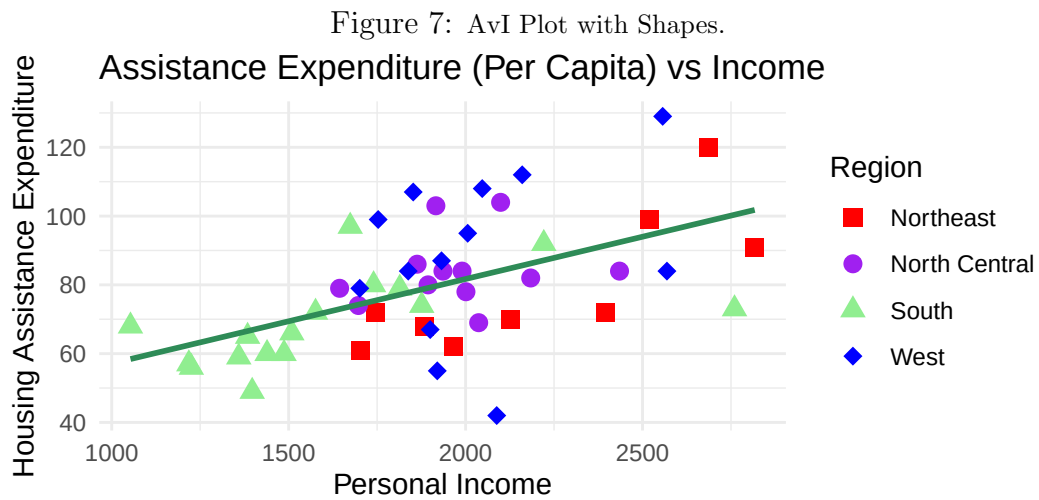
Figure 5: Assistance vs Income Plot.



Next, I added a regression line:



And finally added some fun shapes for the different regions:



We can see that there is a positive correlation between Y and X1 given that the regression line is positively sloped. This is in agreement with the pie chart we saw in the correlation plot above. All this is to say that there appears to be a reasonable positive correlation between Housing Assistance Expenditure and Personal Income by state. The R code for my final plot (AvI Plot with Shapes) can be seen below:

```

1 fac_region <- factor(expenditure$Region)
2
3 ggplot(expenditure, aes(x = X1,
4                           y = Y,
5                           color = fac_region,
6                           shape = fac_region)) +
7   geom_point(size = 3) +
8   geom_smooth(inherit.aes = FALSE,
9               aes(x = X1, y = Y),
10                method = "lm",
11                se = FALSE,
12                color = "seagreen") +
13   scale_color_manual(values = c("red", "purple", "lightgreen", "blue"),
14                      labels = c("1" = "Northeast",
15                                  "2" = "North Central",
16                                  "3" = "South",
17                                  "4" = "West")) +
18   scale_shape_manual(values = c(15, 16, 17, 18),
19                      labels = c("1" = "Northeast",
20                                  "2" = "North Central",
21                                  "3" = "South",
22                                  "4" = "West")) +
23   labs(title = "Assistance Expenditure (Per Capita) vs Income",
24        x = "Personal Income",
25        y = "Housing Assistance Expenditure",
26        color = "Region",
27        shape = "Region") +
28   guides(color = guide_legend("Region"),
29          shape = guide_legend("Region"))

```